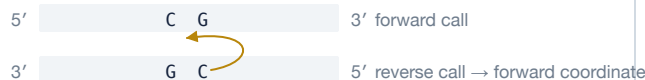


a Per-read CpG state from the ML probability



b CpG strand pooling



M, U, A = methylated, unmethylated, ambiguous reads after pooling

c Site genotype

heterozygous if $\min(M,U) / \max(M,U) \geq 0.6$

and $A / (M+U+A) \leq 0.2$; otherwise homozygous or excluded



■ methylated ■ unmethylated ■ ambiguous

e Output record

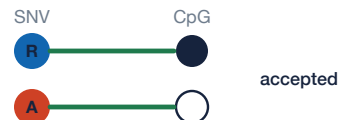
chr1 10469 . N . . PASS RS=P;MR=read3,read7;NR=read1,read4
GT:MD:UD:DP 0/1:14:12:28

MR / NR read names attach the 5mC alleles to reads in phase

d Co-segregation with a neighbouring marker

$$\text{linkage} = \frac{\max(RR + AA, RA + AR)}{RR + AA + RA + AR} \geq 0.9$$

SNV-anchored



SNV within the next 20 variants, on $> \max(6, (d_1+d_2)/4)$ reads

no SNV in range



held for methylation-only mode

CpG-CpG expansion



linked to an accepted CpG; 2 rounds, adjacent CpGs merged

● methylated allele ○ unmethylated allele